tabstat — Compact table of summary statistics

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Description

`tabstat` displays summary statistics for a series of numeric variables in one table. It allows you to specify the list of statistics to be displayed. Statistics can be calculated (conditioned on) another variable. `tabstat` allows substantial flexibility in terms of the statistics presented and the format of the table.

Quick start

Mean of `v1` displayed using `v1`'s display format

```
tabstat v1, format
```

Same as above, but use format with 2 significant digits and a comma

```
tabstat v1, format(%9.2fc)
```

Nonmissing observations, mean, standard error, and coefficient of variation for `v1`

```
tabstat v1, statistics(n mean semean cv)
```

Quartiles and interquartile range of `v1` and `v2`

```
tabstat v1 v2, statistics(q iqr)
```

Same as above, but report statistics separately for each level of `catvar`

```
tabstat v1 v2, by(catvar) statistics(q iqr)
```

Same as above, but display a separate column for each statistic

```
tabstat v1 v2, by(catvar) statistics(q iqr) columns(statistics)
```

Menu

Statistics > Summaries, tables, and tests > Other tables > Compact table of summary statistics

Syntax

```
tabstat varlist [if] [in] [weight] [, options]
```

options	Description
Main	
`by(`*varname*`)`	group statistics by variable
`statistics(`*statname*`[...])`	report specified statistics
Options	
`labelwidth(#)`	width for `by()` variable labels; default is `labelwidth(16)`
`varwidth(#)`	variable width; default is `varwidth(12)`
`columns(variables)`	display variables in table columns; the default
`columns(statistics)`	display statistics in table columns
`format[(%`*fmt*`)]`	display format for statistics; default format is `%9.0g`
`casewise`	perform casewise deletion of observations
`nototal`	do not report overall statistics; use with `by()`
`missing`	report statistics for missing values of `by()` variable
`noseparator`	do not use separator line between `by()` categories
`longstub`	make left table stub wider
`save`	store summary statistics in `r()`

`by` is allowed; see [D] **by**.

`aweights` and `fweights` are allowed; see [U] **11.1.6 weight**.

Options

Main

by(*varname*) specifies that the statistics be displayed separately for each unique value of *varname*; *varname* may be numeric or string. For instance, tabstat height would present the overall mean of height. tabstat height, by(sex) would present the mean height of males, and of females, and the overall mean height. Do not confuse the by() option with the by prefix (see [D] **by**); both may be specified.

statistics(*statname* [...]) specifies the statistics to be displayed; the default is equivalent to specifying statistics(mean). (stats() is a synonym for statistics().) Multiple statistics may be specified and are separated by white space, such as statistics(mean sd). Available statistics are

statname	Definition	*statname*	Definition
mean	mean	p1	1st percentile
count	count of nonmissing observations	p5	5th percentile
n	same as count	p10	10th percentile
sum	sum	p25	25th percentile
max	maximum	median	median (same as p50)
min	minimum	p50	50th percentile (same as median)
range	range = max − min	p75	75th percentile
sd	standard deviation	p90	90th percentile
variance	variance	p95	95th percentile
cv	coefficient of variation (sd/mean)	p99	99th percentile
semean	standard error of mean (sd/$\sqrt{n}$)	iqr	interquartile range = p75 − p25
skewness	skewness	q	equivalent to specifying p25 p50 p75
kurtosis	kurtosis		

Options

labelwidth(#) specifies the maximum width to be used within the stub to display the labels of the by() variable. The default is labelwidth(16). $8 \leq \# \leq 32$.

varwidth(#) specifies the maximum width to be used within the stub to display the names of the variables. The default is varwidth(12). varwidth() is effective only with columns(statistics). Setting varwidth() implies longstub. $8 \leq \# \leq 32$.

columns(variables | statistics) specifies whether to display variables or statistics in the columns of the table. columns(variables) is the default when more than one variable is specified.

format and format(%*fmt*) specify how the statistics are to be formatted. The default is to use a %9.0g format.

format specifies that each variable's statistics be formatted with the variable's display format; see [D] **format**.

format(%*fmt*) specifies the format to be used for all statistics.

The column width is the maximum width of these formats. The minimum column width is nine display characters.

casewise specifies casewise deletion of observations. Statistics are to be computed for the sample that is not missing for any of the variables in *varlist*. The default is to use all the nonmissing values for each variable.

nototal is for use with by(); it specifies that the overall statistics not be reported.

missing specifies that missing values of the by() variable be treated just like any other value and that statistics should be displayed for them. The default is not to report the statistics for the by()== *missing* group. If the by() variable is a string variable, by()=="" is considered to mean missing.

noseparator specifies that a separator line between the by() categories not be displayed.

longstub specifies that the left stub of the table be made wider so that it can include names of the statistics or variables in addition to the categories of by(*varname*). The default is to describe the statistics or variables in a header. longstub is ignored if by(*varname*) is not specified.

save specifies that the summary statistics be returned in r(). The overall (unconditional) statistics are returned in matrix r(StatTotal) (rows are statistics, columns are variables). The conditional statistics are returned in the matrices r(Stat1), r(Stat2), ..., and the names of the corresponding variables are returned in the macros r(name1), r(name2), ...

Remarks and examples

This command is probably most easily understood by going through a series of examples.

▷ Example 1

We have data on the price, weight, mileage rating, and repair record of 22 foreign and 52 domestic 1978 automobiles. We want to summarize these variables for the different origins of the automobiles.

```
. use https://www.stata-press.com/data/r19/auto
(1978 automobile data)
. tabstat price weight mpg rep78, by(foreign)
Summary statistics: Mean
Group variable: foreign (Car origin)
```

foreign	price	weight	mpg	rep78
Domestic	6072.423	3317.115	19.82692	3.020833
Foreign	6384.682	2315.909	24.77273	4.285714
Total	6165.257	3019.459	21.2973	3.405797

More summary statistics can be requested via the `statistics()` option. The group totals can be suppressed with the `nototal` option.

```
. tabstat price weight mpg rep78, by(foreign) stat(mean sd min max) nototal
Summary statistics: Mean, SD, Min, Max
Group variable: foreign (Car origin)
```

foreign	price	weight	mpg	rep78
Domestic	6072.423	3317.115	19.82692	3.020833
	3097.104	695.3637	4.743297	.837666
	3291	1800	12	1
	15906	4840	34	5
Foreign	6384.682	2315.909	24.77273	4.285714
	2621.915	433.0035	6.611187	.7171372
	3748	1760	14	3
	12990	3420	41	5

Although the header of the table describes the statistics running vertically in the "cells", the table may become hard to read, especially with many variables or statistics. The `longstub` option specifies that a column be added describing the contents of the cells. The `format` option can be issued to specify that `tabstat` display the statistics by using the display format of the variables rather than the overall default `%9.0g`.

```
. tabstat price weight mpg rep78, by(foreign) stat(mean sd min max) long format
```

foreign	Stats	price	weight	mpg	rep78
Domestic	Mean	6,072.4	3,317.1	19.8269	3.02083
	SD	3,097.1	695.364	4.7433	.837666
	Min	3,291	1,800	12	1
	Max	15,906	4,840	34	5
Foreign	Mean	6,384.7	2,315.9	24.7727	4.28571
	SD	2,621.9	433.003	6.61119	.717137
	Min	3,748	1,760	14	3
	Max	12,990	3,420	41	5
Total	Mean	6,165.3	3,019.5	21.2973	3.4058
	SD	2,949.5	777.194	5.7855	.989932
	Min	3,291	1,760	12	1
	Max	15,906	4,840	41	5

We can specify a layout of the table in which the statistics run horizontally and the variables run vertically by specifying the col(statistics) option.

```
. tabstat price weight mpg rep78, by(foreign) stat(min mean max) col(stat) long
```

foreign	Variable	Min	Mean	Max
Domestic	price	3291	6072.423	15906
	weight	1800	3317.115	4840
	mpg	12	19.82692	34
	rep78	1	3.020833	5
Foreign	price	3748	6384.682	12990
	weight	1760	2315.909	3420
	mpg	14	24.77273	41
	rep78	3	4.285714	5
Total	price	3291	6165.257	15906
	weight	1760	3019.459	4840
	mpg	12	21.2973	41
	rep78	1	3.405797	5

Finally, tabstat can also be used to enhance summarize so we can specify the statistics to be displayed. For instance, we can display the number of observations, the mean, the coefficient of variation, and the 25%, 50%, and 75% quantiles for a list of variables.

```
. tabstat price weight mpg rep78, stat(n mean cv q) col(stat)
```

variable	N	mean	cv	p25	p50	p75
price	74	6165.257	.478406	4195	5006.5	6342
weight	74	3019.459	.2573949	2240	3190	3600
mpg	74	21.2973	.2716543	18	20	25
rep78	69	3.405797	.290661	3	3	4

Because we did not specify the by() option, these statistics were not displayed for the subgroups of the data formed by the categories of the by() variable.

◁

Video example

Descriptive statistics in Stata

Acknowledgments

The tabstat command was written by Jeroen Weesie and Vincent Buskens both of the Department of Sociology at Utrecht University, The Netherlands.

Reference

Donath, S. 2018. baselinetable: A command for creating one- and two-way tables of summary statistics. *Stata Journal* 18: 327–344.

Also see

[R] **summarize** — Summary statistics

[R] **table** — Table of frequencies, summaries, and command results

[R] **table summary** — Table of summary statistics

[R] **tabulate, summarize()** — One- and two-way tables of summary statistics

[D] **collapse** — Make dataset of summary statistics

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